MAT2540, Classwork7, Spring2026

6.2 *The Pigeonhole Principle*

1. Theorem: *The Pigeonhole Principle* P Q

If k is a positive integer and $k+1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

Using a Proof By contraposition: Suppose that none of k boxes contains more than 1 object. Then the total number of objects would be at most k. This is a contradiction, because there are at least $k+1$ objects

2. Corollary: A function f from a set with $k+1$ or more elements to a set with k elements is not one-to-one.

one-to-one: $f(x)=y$ (input, output) and f is 1-1

$\Rightarrow$ same y gets the same x

domain $\Rightarrow$ objects codomain $\rightarrow$ boxes

x_1, x_2, x_3, ..., x_k, x_{k+1} $\rightarrow$ y_1, y_2, y_3, ..., y_k k boxes

$f(x_1)=y_1$

$f(x_2)=y_2$ and $f(x_{k+1})=y_2$

By the Pigeonhole principle, if we put all $k+1$ of x into k boxes, then one of them contains 2 or more x and it means f is NOT 1-to-one.

3. How many people in a group do you need if there is at least two with the same birthday?

By the Pigeonhole principle, since there are 366 birthdays (366 boxes), then we need $366+1=367$ people in order to have at least two ppl with the same birthday.

4. Find a multiple of 7 which has only 0s and 1s in its decimal expansion.

number	1	11	111	1111	11111	111111	1111111	11111111
remainder	1	4	6	5	2	0	1	4

$7\cdot q_2+1$ $7\cdot q_1+1$

$1111111-1=1111110$

5. Show that for every integer n there is a multiple of n that has only 0s and 1s in its decimal expansion.

Let n be positive integer. Consider the $n+1$ integers $1, 11, 111, \ldots, \underbrace{1\cdots1}_{n+1 \text{ of } 1s}$.
Note that there are n possible remainders when an integer is divided by n.
By the Pigeonhole Priciple, there must be two out of $n+1$ with the same remainder when divided by n, denoted by a, b, $a<b$
Then $b-a$ will be a multiple of n that has only 0s and 1s.

6. Theorem: *The Generalized Pigeonhole Principle* P Q $P\rightarrow Q \Rightarrow \sim Q \rightarrow \sim P$

If N objects are placed into k boxes, then there is at least one box containing at least $\left\lceil \frac{N}{k} \right\rceil$ objects.

A proof by contraposition. Suppose that none of boxes contains more than $\lceil \frac{N}{k} \rceil - 1$ objects

Then, the total number of objects is at most

$$\boxed{k\cdot\left(\left\lceil \frac{N}{k}\right\rceil - 1\right)} < k\cdot\left(\frac{N}{k}+1-1\right) = k\cdot\frac{N}{k} = N$$

where $\lceil \frac{N}{k} \rceil < \frac{N}{k}+1$. Thus, the total number of objects is less than N which is a contradition

7. What is the **minimum number** of objects such that at least r of these objects must be in one of k boxes when these objects are distributed among the boxes?

Let N be the minimum number of objects. We have $\lceil \frac{N}{k} \rceil \geq r \Rightarrow$

$\boxed{\frac{N}{k}+1 > \lceil \frac{N}{k} \rceil \geq r} \Rightarrow \frac{N}{k}+1 > r \Rightarrow \frac{N}{k} > r-1 \Rightarrow N > k(r-1)$

and the smallest $N = k\cdot(r-1)+1$

8. a) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are selected?

boxes $\Rightarrow k=4$ objects $r=3$

N = The minimum number of card selected is $4\times(3-1)+1 = 9$.

(What happens if there are only 8 cards selected?)

b) How many must be selected from a standard deck of 52 cards to guarantee that at least three hearts are selected?

$13+13+13+3=42$

9. Show that if five integers are selected from the first eight positive integers, there must be a pair of these integers with a sum equal to 9. $\{1,2,3,4,5,6,7,8\}$

The pairs of subset that sum as 9: $\{1,8\}, \{2,7\}, \{3,6\}, \{4,5\}$.

If five integers are selected from 1 to 8, by Pigeohole principle at least two of them from the same subset.

10. During a month with 30 days, a baseball team plays at least one game a day, but no more than 45 games. Show that there must be a period of some number of consecutive days during which the team must play exactly 14 games.

Let a_j be the number of the games played on or before the jth day.

The sequence $a_1, \cdots, a_{30}$ is a strictly increasing sequence, with

$1 \leq a_j \leq 45$.

Moreover, $a_1+14, a_2+14, a_3+14, \cdots, a_{30}+14$ is also a strictly increasing sequence, $15 \leq a_j+14 \leq 59$.

The 60 positive integer, $a_1, a_2, \cdots, a_{30}, a_1+14, a_2+14, a_3+14, \cdots, a_{30}+14$ are all less than or equal to 59

Because a_j, $j=1,\cdots,30$ are distinct and a_i+14, $i=1,\cdots,30$ are distinct, thus, there exists an i, j s.t. $a_j = a_i+14$

which means that exactly 14 games were played beteen $i+1$th to jth day.